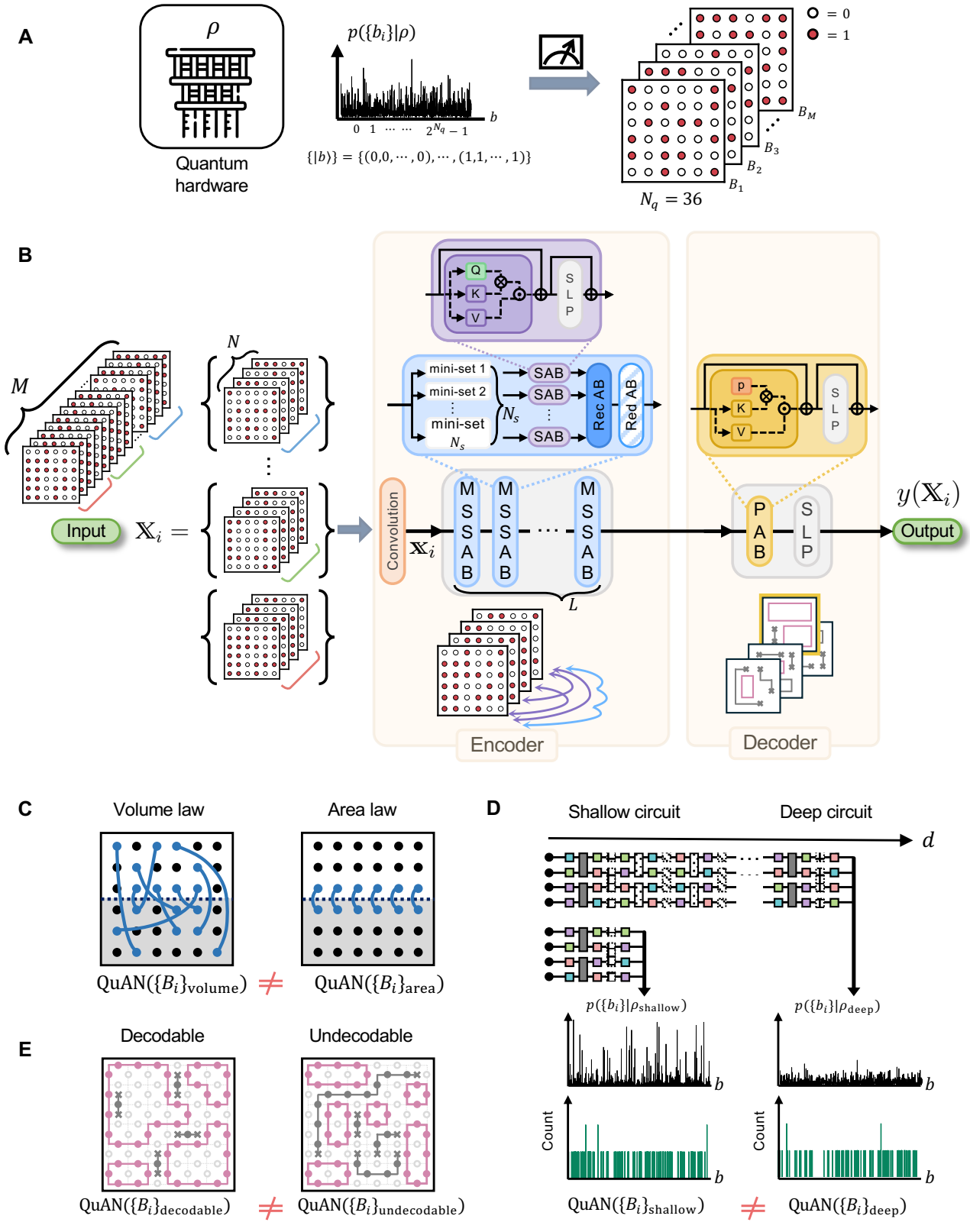
A
ρ
Quantum hardware
$p(\{b_i\}|\rho)$
b
0 1 ··· ··· $2^{N_q}-1$
$\{|b\rangle\} = \{(0,0,\cdots,0),\cdots,(1,1,\cdots,1)\}$
= 0
= 1
B_M
B_3
B_2
B_1
$N_q = 36$
B
M
N
Input
$\mathbb{X}_i =$
Convolution
$\mathbb{X}_i$
MSSAB
L
mini-set 1
mini-set 2
mini-set N_s
SAB
Rec AB
Red AB
Q
K
V
SLP
p
PAB
$y(\mathbb{X}_i)$
Output
Encoder
Decoder
C
Volume law
Area law
$\mathrm{QuAN}(\{B_i\}_{\mathrm{volume}}) \neq \mathrm{QuAN}(\{B_i\}_{\mathrm{area}})$
D
Shallow circuit
Deep circuit
d
$p(\{b_i\}|\rho_{\mathrm{shallow}})$
$p(\{b_i\}|\rho_{\mathrm{deep}})$
b
Count
$\mathrm{QuAN}(\{B_i\}_{\mathrm{shallow}}) \neq \mathrm{QuAN}(\{B_i\}_{\mathrm{deep}})$
E
Decodable
Undecodable
$\mathrm{QuAN}(\{B_i\}_{\mathrm{decodable}}) \neq \mathrm{QuAN}(\{B_i\}_{\mathrm{undecodable}})$